DESIGN OF TEST CASES AND MAINTENANCE PLAN FOR AN E-COMMERCE PLATFORM

AIM

To design comprehensive test cases and develop an effective maintenance plan for an E-Commerce Platform to ensure its functionality, reliability, security, performance, and long-term operational efficiency.

TEST CASES

The following test cases ensure that the major functionalities of the e-commerce platform work correctly and meet user requirements.

TEST CASE ID	TEST SCENARIO	TEST STEPS	EXPECTED RESULT	STATUS
TC-01	User Registration	Enter valid user details and click **Register**	User account is created successfully	Pass
TC-02	User Login	Enter valid username and password	User is logged into the application	Pass
TC-03	Invalid login	Enter incorrect username or password	Error message "Invalid Credentials" is displayed	Pass
TC-04	Product search	Search for a product using keywords	Relevant products are displayed	Pass
TC-05	Add to Cart	Select a product and click **Add to Cart**	Product is added to the shopping cart	Pass
TC-06	Place order	Proceed to checkout and confirm order	Order is placed successfully with an Order ID	Pass
TC-07	Online payment	complete payment using a valid payment method	Payment is processed successfully	Pass
TC-08	Order Tracking	View order	Current order	Pass

		status from the customer dashboard	status is displayed correctly	
TC-09	Inventory Update	Purchase the order	Product stock is reduced automatically	Pass
TC-10	Admin Product Management	Add or update a product in the admin panel	Product details are saved successfully	Pass

TYPES OF TESTING PERFORMED

- Unit Testing
- Integration Testing
- System Testing
- Functional Testing
- Performance Testing
- Security Testing
- User Acceptance Testing (UAT)

MAINTENANCE PLAN

Corrective Maintenance

- Fix bugs reported by users.
- Resolve payment and order issues.
- Correct system and database errors.

Adaptive Maintenance

- Update system for new browsers and OS.
- Integrate new payment gateways/services.
- Adjust to changing business needs.

Perfective Maintenance

- Improve performance and loading speed.
- Enhance UI/UX experience

Preventive Maintenance

- Apply security updates and patches.
- Perform regular database backups.
- Monitor system health and performance.

EXPECTED OUTCOMES

- Reliable and error-free application performance.
- Secure customer data and online transactions.
- Faster response time and improved user experience.
- Reduced system downtime.
- Continuous feature improvements based on customer feedback.
- Increased customer satisfaction and business growth.

CONCLUSION

The design of test cases and a maintenance plan ensures that the E-Commerce Platform functions correctly, securely, and efficiently. Thorough testing helps identify and resolve defects before deployment, while regular maintenance improves system performance, enhances security, and supports new feature updates. Together, these activities ensure reliable software, high customer satisfaction, and the long-term success of the e-commerce platform.